



نظم التشغيل

12:00-2:00

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Course code: CS321

3rd Level

Midterm Exam

Duration: 2 hours

Total marker: 50

Reviewer: (Add the name)

This exam for the following program(s) : (Add the program's name(s))

* Required

* This form will record your name, please fill your name.

1

* الإسم الرباعي (بالعربي فقط)

2

* رقم الجلوس

3

* المستوي

- ☐ الاول
- ☐ الثاني
- ☐ الثالث
- ☐ الرابع 2013
- ☐ الرابع 2014
- ☐ الرابع 2015
- ☐ الرابع 2016
- ☐ الرابع 2017
- ☐ الرابع 2018

4

* البرنامج

- ☐ عام
- ☐ بايو
- ☐ هندسة

5

* رقم المعمل

- ☐ ج٠
- ☐ د٠
- ☐ ه٠
- ☐ اب
- ☐ اج
- ☐ اد
- ☐ اه
- ☐ أ٢
- ☐ ب٢
- ☐ ج٢
- ☐ د٢
- ☐ ه٢
- ☐ أ٣
- ☐ ب٣

- ☐ ج^٣
- ☐ د^٣
- ☐ هـ^٣
- ☐ أ^٤
- ☐ ب^٤
- ☐ ج^٤
- ☐ د^٤
- ☐ هـ^٤

6

* رقم الكمبيوتر

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

8

An I/O device that uses DMA I/O has no need to interrupt the processor
(1 Point)

- ☐ True
- ☒ False

9

If a process must loop repeatedly, keeping the CPU occupied, to wait for a condition, that is called (1 Point)

- ☐ starvation
- ☐ deadlock
- ☒ busy waiting
- ☐ mutual exclusion

10

The condition that a process may hold a resource while awaiting assignment of additional resources is called (1 Point)

- ☐ "transitive wait" condition
- ☒ "hold and wait" condition
- ☐ "mutual exclusion" condition
- ☐ none of the mentioned

11

The condition that a resource cannot be forcibly taken away from a process that is holding it is called (1 Point)

- ☐ "blocking"
- ☐ "preemption"
- ☐ "mutual exclusion"
- ☒ "no preemption"

12

If resources are assigned numbers and required to be requested in order, so that a process cannot hold a higher-numbered resource while it requests a lowered number resource, deadlock is not possible. (1 Point)

- ☒ True
- ☐ False

13

Once a system is in an unsafe state eventual deadlock is certain (1 Point)

- ☐ True
- ☒ False

14

Shifting the regions of memory allocated to processes, so that all the free memory forms one contiguous region is called..... (1 Point)

- ☐ internal fragmentation
- ☐ placement
- ☒ compaction
- ☐ swapping

15

Paging eliminates the problem of external fragmentation (1 Point)

- ☒ True
- ☐ False

16

Given a paged memory system with page size 1024, the offset of logical address 2500 is....., all numbers are in decimal, everything is numbered starting from zero. (1 Point)

- ☐ 448
- ☐ 1476
- ☒ 452
- ☐ 500

17

The first-fit algorithm searches the entire list of free blocks (1 Point)

- ☐ True
- ☒ False

18

In the Banker's algorithm, the content of the matrix Need is (1 Point)

- ☒ Max – Allocation
- ☐ Allocation – Available
- ☐ Max – Available
- ☐ Allocation – Max

19

Which of the following takes the least amount of time to handle? (1 Point)

- ☐ disk read operation
- ☐ cache miss
- ☐ page fault
- ☒ TLB miss

20

The term "aging" is sometimes used for the policy of promoting a process to a higher-priority queue (in a feedback queue scheduling scheme) after it spends a certain amount of time waiting for service in its current queue. The purpose of this policy is to: (1 Point)

- ☐ prevent deadlock
- ☐ raise the average multiprogramming level
- ☐ increase average throughput
- ☐ shorten average response time
- ☐ prevent starvation

21

Which if the following is **not** one of the motivations for implementing virtual memory? (1 Point)

- ☐ the processor may be used more efficiently, due to a higher degree of multiprogramming
- ☐ a process may be larger than all of main memory
- ☐ the implementation of memory management is simplified
- ☐ more processes may be maintained in main memory

22

virtual page number	valid bit	frame number
0	v	3
1	v	4
2	i	-
3	v	2
4	v	1
5	i	-

Suppose the page table for the process currently executing on the processor contains the following data, where all numbers are in decimal, everything is numbered starting from zero, all addresses are memory byte addresses, and the page size is 1024 bytes. What physical address, if any, would the virtual address 2050 correspond to? (1 Point)

- ☐ 2050
- ☐ 1026
- ☐ 4098
- ☐ none; a page fault would occur

23

The number of processes completed per unit of time is a measure of (1 Point)

- ☐ fairness
- ☐ throughput
- ☐ processor utilization
- ☐ predictability

24

For which of the following CPU scheduling policies is starvation possible?

(1 Point)

- ☐ Round Robin
- ☒ Priority
- ☐ FCFS

25

Suppose a series of disk I/O requests has been made, for tracks 125, 25, 150, and 50 (in that order), and the head is currently serving a request at track 60, having moved there from track 10. What is the order the four pending requests would be served by the SCAN algorithm? (1 Point)

- ☐ 125, 25, 150, 50
- ☐ 25, 50, 125, 150
- ☒ 125, 150, 50, 25
- ☐ 50, 25, 125, 150
- ☐ 125, 150, 25, 50

26

Suppose a series of disk I/O requests has been made, for tracks 125, 25, 150, and 50 (in that order), and the head is currently serving a request at track 60, having moved there from track 10. What is the order the four pending requests would be served by the C-SCAN algorithm? (1 Point)

- ☐ 50, 25, 125, 150
- ☐ 125, 150, 50, 25
- ☒ 125, 150, 25, 50
- ☐ 125, 25, 150, 50
- ☐ 25, 50, 125, 150

27

Suppose a series of disk I/O requests has been made, for tracks 125, 25, 150, and 50 (in that order), and the head is currently serving a request at track 60, having moved there from track 10. What is the order the four pending requests would be served by the SSTF algorithm? (1 Point)

- ☐ 125, 150, 50, 25
- ☐ 25, 50, 125, 150
- ☐ 125, 150, 25, 50
- ☒ 50, 25, 125, 150
- ☐ 125, 25, 150, 50

28

Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process Burst Time

P1	6
P2	1
P3	3
P4	2
P5	5

The processes are assumed to have arrived all at time 0, at order P1, P2, P3, P4, and P5.

What is the turnaround time of process P3 using the Shortest Job First (SJF) algorithm? (1 Point)

☐ 6

☐ 12

☐ 3

☐ 2

☐ 0

29

Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process Burst Time

P1	6
P2	1
P3	3
P4	2
P5	5

The processes are assumed to have arrived all at time 0, at order P1, P2, P3, P4, and P5.

What is waiting time of process P3 using the Shortest Job First (SJF) algorithm?
(1 Point)

- ☒ 3
- ☐ 12
- ☐ 2
- ☐ 6
- ☐ 0

30

Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process Burst Time

P1	6
P2	1
P3	3
P4	2
P5	5

The processes are assumed to have arrived all at time 0, at order P1, P2, P3, P4, and P5.

What is waiting time of process P3 using the Round Robin (RR, Quantum = 2 msec) algorithm? (1 Point)

☐ 2

☐ 6

☒ 9

☐ 3

☐ 12

☐ 0

31

Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process Burst Time

P1	6
P2	1
P3	3
P4	2
P5	5

The processes are assumed to have arrived all at time 0, at order P1, P2, P3, P4, and P5.

Which of the algorithms results in the minimal average waiting time? (1 Point)

☐ RR, Quantum = 2 msec

☒ SIF

32

What would be the initial value of semaphore "S" when it is used for Mutual Exclusion: (1 Point)

☐ 0

☐ -1

☒ 1

☐ 2

33

What would be the initial value of semaphore "S" when it is used for resource allocation for five resources: (1 Point)

- ☐ 1
- ☒ 5
- ☐ 0
- ☐ -1

34

Below the structure of the Readers & Writers problem is shown using the semaphores mutex & wrt and an integer variable readCount.

READER

```
wait(mutex);
readCount++;
if (readCount == 1)
    wait(wrt);
signal(mutex);
...
reading section
...
wait(mutex);
readCount--;
if (readCount == 0)
    signal(wrt);
signal(mutex);
```

WRITER

```
wait(wrt);
.....
writing section
.....
signal(wrt);
```

In the above algorithm what are the initial values of *mutex*, *wrt*, and *readCount*?
(1 Point)

- ☐ mutex = 1, wrt = 1, readCount = 0
- ☐ mutex = 1, wrt = 0, readCount = 0
- ☒ mutex = 1, wrt = 1, readCount = 1
- ☐ mutex = 0, wrt = 1, readCount = 0

35

In the readers/writers problem we want to allow multiple readers to be in their critical sections for the shared data at the same time (1 Point)

- ☒ True
- ☐ False

36

In the readers/writers problem, we want to allow a reader and a writer to be in their critical sections for the shared data at the same time. (1 Point)

- ☐ True
- ☒ False

37

Critical section is a part of a program where two or more processes access a shared resource (1 Point)

- ☒ True
- ☐ False

38

Consider a demand-paging system with the following time-measured utilizations:

CPU utilization 18%

Paging disk 96%

Other I/O devices 6%

Install a bigger paging disk will improve CPU utilization. (1 Point)

☐ True

☒ False

39

Consider a demand-paging system with the following time-measured utilizations:

CPU utilization 18%

Paging disk 96%

Other I/O devices 6%

Decrease the degree of multiprogramming will improve CPU utilization.

(1 Point)

☒ True

☐ False

40

Consider the following page-reference strings:

1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

How many page faults would occur using LRU replacement algorithm, assuming four memory frames? (1 Point)

- ☐ 6
- ☐ 8
- ☒ 10
- ☐ 12
- ☐ 14
- ☐ 15

41

Consider the following page-reference strings:

1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

How many page faults would occur using FIFO replacement algorithm, assuming four memory frames? (1 Point)

- ☐ 6
- ☐ 8
- ☐ 10
- ☐ 12
- ☒ 14
- ☐ 15

42

Consider the following page-reference strings:

1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

How many page faults would occur using optimal algorithm, assuming four memory frames? (1 Point)

- ☐ 6
- ☒ 8
- ☐ 10
- ☐ 12
- ☐ 14
- ☐ 15

43

Consider a file currently consisting of 100 blocks. Assume that the file control block is already in memory. Calculate how many disk I/O operations are required for linked allocation strategy for adding one block at the end. Assume that the block information to be added is stored in memory. (1 Point)

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 100
- ☒ 102

44

Consider a file currently consisting of 100 blocks. Assume that the file control block is already in memory. Calculate how many disk I/O operations are required for linked allocation strategy for removing one block from the end. Assume that the block information to be added is stored in memory. (1 Point)

- ☐ 1
- ☐ 2
- ☐ 3
- ☒ 100
- ☐ 102

45

What is a short-term scheduler? (1 Point)

- ☐ It selects which process must be brought into the ready queue
- ☒ It selects which process must be executed next and allocates CPU
- ☐ It selects which process must be removed from memory by swapping
- ☐ None of the mentioned

46

Suppose that a process is in "Blocked" state waiting for some I/O service. When the service is completed, it goes to the _____ (1 Point)

- ☐ Suspended state
- ☒ Ready state
- ☐ Terminated state
- ☐ Running state

47

The real difficulty with SJF in short term scheduling is _____ (1 Point)

- ☐ it is too good an algorithm
- ☒ knowing the length of the next CPU request
- ☐ it is too complex to understand
- ☐ none of the mentioned

48

The address generated by the CPU is referred to as _____ (1 Point)

- ☐ Physical address
- ☒ Logical address
- ☐ None of the mentioned

49

The first fit, best fit and worst fit are strategies to select a _____ (1 Point)

- ☐ process from a queue to put in memory
- ☐ processor to run the next process
- ☐ free hole from a set of available holes
- ☐ all of the mentioned

50

Disk request scheduling algorithms attempt to minimize (1 Point)

- ☐ Seek time
- ☐ Transfer time
- ☐ Rotational latency

51

The page replacement policy that sometimes leads to more page faults when the size of the memory is increased is: (1 Point)

- ☐ least-recently-used (LRU) page replacement
- ☐ Optimal page replacement
- ☐ BEST page replacement
- ☐ FIFO page replacement

52

Dijkstra's banking algorithm solves the problem of (1 Point)

- ☐ deadlock recovery
- ☐ mutual exclusion
- ☒ deadlock avoidance
- ☐ context switching

53

Dirty bit is used to show the (1 Point)

- ☒ page that is modified after being loaded into cache memory
- ☐ page that is less frequently accessed
- ☐ page with corrupted data
- ☐ wrong page in the memory

54

The best-fit algorithm can be used for: (1 Point)

- ☐ FAT allocation
- ☒ contiguous allocation
- ☐ indexed allocation
- ☐ linked allocation

55

If there are 32 pages, each of size 1 Kbytes, then the logical address should have (1 Point)

- ☐ 14 bits
- ☐ 16 bits
- ☒ 15 bits
- ☐ 10 bits

56

In paged memory systems, if the page size is increased, then the internal fragmentation generally (1 Point)

- ☒ becomes more
- ☐ becomes less
- ☐ has no effect
- ☐ remains constant

57

Disk scheduling involves deciding (1 Point)

- ☒ the order in which disk access requests must be serviced
- ☐ which file should be accessed next
- ☐ the physical location where files should be accessed in the disk
- ☐ which disk should be accessed next

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